

Wave 2C Residual Harmonic Temporal Hybrid Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved wave 2C residual harmonic temporal hybrid campaign. The campaign tested residual harmonic GRU and residual harmonic LSTM sequence regressors across global, Fw, and Bw direction surfaces, with three harmonic-basis tiers per surface: sparse RCIM, dense 0..240, and dense 0..360.

The campaign completed all 18 planned runs with zero launcher failures. The campaign winner by the configured selection policy, minimum test_mae with test_rmse, val_mae, and trainable parameter count as tie breakers, is te_residual_harmonic_gru_sequence_remote_Fw_sparse_rcim.

Campaign Artifacts

Artifact	Path
Campaign output	output/training_campaigns/2026-05-27-18-55-47_wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27
Campaign leaderboard	output/training_campaigns/2026-05-27-18-55-47_wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27/campaign_leaderboard.yaml
Best-run pointer	output/training_campaigns/2026-05-27-18-55-47_wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27/campaign_best_run.yaml
Execution report	output/training_campaigns/2026-05-27-18-55-47_wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27/campaign_execution_report.md
Planning report	doc/reports/campaign_plans/wave2/2026-05-27-18-08-32_wave2c_residual_harmonic_temporal_hybrid_campaign_plan_report.md
Model technical document	doc/technical/2026-05/2026-05-27/2026-05-27-18-08-32_wave2c_residual_harmonic_temporal_hybrids.md
Remote launcher standard	doc/technical/2026-05/2026-05-27/2026-05-27-18-35-06_campaign_launcher_remote_execution_standard.md
Closeout technical document	doc/technical/2026-05/2026-05-28/2026-05-28-11-35-34_wave2c_campaign_closeout.md

Execution Summary

Field	Value
Campaign name	wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27
Started at	2026-05-27T18:55:47
Finished at	2026-05-27T22:35:20
Completed runs	18
Failed runs	0
Tested model types	residual_harmonic_gru_sequence , residual_harmonic_lstm_sequence
Tested direction scopes	global , Fw , Bw
Harmonic basis tiers	sparse_rcim , dense_240 , dense_360
Sparse harmonic list	[0, 1, 3, 39, 40, 78, 81, 156, 162, 240]

Campaign Winner

Field	Value
Run name	te_residual_harmonic_gru_sequence_remote_fw_sparse_rcim
Run instance	2026-05-27-19-07-31_te_residual_harmonic_gru_sequence_remote_fw_sparse_rcim
Model family	residual_harmonic_gru_sequence_fw_sparse_rcim
Model type	residual_harmonic_gru_sequence
Direction scope	forward only
Trainable parameters	151,060
Validation MAE	0.003309
Validation RMSE	0.003828
Test MAE	0.003200
Test RMSE	0.003635

The winning checkpoint is stored at
output/training_runs/residual_harmonic_gru_sequence_fw_sparse_rcim/2026-05-27-19-07-31_te_residual_harmonic_gru_sequence_remote_fw_sparse_rcim/checkpoints/residual_harmonic_gru_sequence-epoch=015-val_mae=0.00330934.ckpt .

Leaderboard

Rank	Run	Family	Scope	Test MAE	Test RMSE	Val MAE	Params
1	te_residual_harmonic_gru_sequence_remote_Fw_sparse_rcim	residual_harmonic_gru_sequence_fw_sparse_rcim	Fw	0.003200	0.003635	0.003309	151,060
2	te_residual_harmonic_gru_sequence_remote_Fw_dense240	residual_harmonic_gru_sequence_fw_dense240	Fw	0.003219	0.003653	0.003270	151,522
3	te_residual_harmonic_lstm_sequence_remote_Fw_sparse_rcim	residual_harmonic_lstm_sequence_fw_sparse_rcim	Fw	0.003234	0.003679	0.003344	201,364
4	te_residual_harmonic_gru_sequence_remote_Fw_dense360	residual_harmonic_gru_sequence_fw_dense360	Fw	0.003241	0.003677	0.003265	151,762
5	te_residual_harmonic_lstm_sequence_remote_Fw_dense240	residual_harmonic_lstm_sequence_fw_dense240	Fw	0.003262	0.003706	0.003307	201,826
6	te_residual_harmonic_lstm_sequence_remote_Fw_dense360	residual_harmonic_lstm_sequence_fw_dense360	Fw	0.003351	0.003774	0.003302	202,066
7	te_residual_harmonic_lstm_sequence_remote_global_sparse_rcim	residual_harmonic_lstm_sequence_sparse_rcim	global	0.003368	0.003808	0.003632	201,364
8	te_residual_harmonic_lstm_sequence_remote_Bw_sparse_rcim	residual_harmonic_lstm_sequence_bw_sparse_rcim	Bw	0.003440	0.004030	0.003764	201,364
9	te_residual_harmonic_gru_sequence_remote_global_sparse_rcim	residual_harmonic_gru_sequence_sparse_rcim	global	0.003440	0.003848	0.003607	151,060
10	te_residual_harmonic_gru_sequence_remote_Bw_dense360	residual_harmonic_gru_sequence_bw_dense360	Bw	0.003468	0.004050	0.003773	151,762
11	te_residual_harmonic_lstm_sequence_remote_global_dense240	residual_harmonic_lstm_sequence_dense240	global	0.003473	0.003925	0.003624	201,826
12	te_residual_harmonic_lstm_sequence_remote_global_dense360	residual_harmonic_lstm_sequence_dense360	global	0.003477	0.003940	0.003648	202,066
13	te_residual_harmonic_gru_sequence_remote_Bw_dense240	residual_harmonic_gru_sequence_bw_dense240	Bw	0.003492	0.004074	0.003585	151,522
14	te_residual_harmonic_gru_sequence_remote_Bw_sparse_rcim	residual_harmonic_gru_sequence_bw_sparse_rcim	Bw	0.003502	0.004061	0.003833	151,060
15	te_residual_harmonic_gru_sequence_remote_global_dense240	residual_harmonic_gru_sequence_dense240	global	0.003511	0.003983	0.003600	151,522
16	te_residual_harmonic_gru_sequence_remote_global_dense360	residual_harmonic_gru_sequence_dense360	global	0.003535	0.003999	0.003628	151,762
17	te_residual_harmonic_lstm_sequence_remote_Bw_dense360	residual_harmonic_lstm_sequence_bw_dense360	Bw	0.003556	0.004125	0.003729	202,066
18	te_residual_harmonic_lstm_sequence_remote_Bw_dense240	residual_harmonic_lstm_sequence_bw_dense240	Bw	0.003605	0.004129	0.003742	201,826

Technical Interpretation

The strongest Wave 2C scalar training result is the forward-only sparse RCIM residual harmonic GRU , with test MAE 0.003200. The dense harmonic branches did not improve the campaign winner: the second ranked model is the forward-only dense 0. .240 residual harmonic GRU , and the third ranked model is the forward-only sparse RCIM residual harmonic LSTM .

This result is execution-valid and useful as a model-family probe, but it does not exceed the current Wave 2B scalar training winner. The program-level scalar best remains te_periodic_gru_sequence_remote_Bw with test MAE 0.002344. That means Wave 2C should be retained as a completed comparison branch, while Wave 2B remains the stronger scalar training baseline until a future campaign or official Track 2 review changes that conclusion.

The campaign also shows that adding a recurrent residual branch over an explicit harmonic base is not automatically superior to feeding the harmonic prior directly into the recurrent temporal model. That is a useful negative result: the residual decomposition remains inspectable, but the sparse periodic recurrent Wave 2B formulation is still the better scalar candidate.

Registry Effects

Registry Scope	Current Family Best	Test MAE	Interpretation
residual_harmonic_gru_sequence_sparse_rcim	te_residual_harmonic_gru_sequence_remote_global_sparse_rcim	0.003440	Updated by this campaign
residual_harmonic_gru_sequence_fw_sparse_rcim	te_residual_harmonic_gru_sequence_remote_Fw_sparse_rcim	0.003200	Updated by this campaign
residual_harmonic_gru_sequence_bw_sparse_rcim	te_residual_harmonic_gru_sequence_remote_Bw_sparse_rcim	0.003502	Updated by this campaign
residual_harmonic_gru_sequence_dense240	te_residual_harmonic_gru_sequence_remote_global_dense240	0.003511	Updated by this campaign
residual_harmonic_gru_sequence_fw_dense240	te_residual_harmonic_gru_sequence_remote_Fw_dense240	0.003219	Updated by this campaign
residual_harmonic_gru_sequence_bw_dense240	te_residual_harmonic_gru_sequence_remote_Bw_dense240	0.003492	Updated by this campaign
residual_harmonic_gru_sequence_dense360	te_residual_harmonic_gru_sequence_remote_global_dense360	0.003535	Updated by this campaign
residual_harmonic_gru_sequence_fw_dense360	te_residual_harmonic_gru_sequence_remote_Fw_dense360	0.003241	Updated by this campaign
residual_harmonic_gru_sequence_bw_dense360	te_residual_harmonic_gru_sequence_remote_Bw_dense360	0.003468	Updated by this campaign
residual_harmonic_lstm_sequence_sparse_rcim	te_residual_harmonic_lstm_sequence_remote_global_sparse_rcim	0.003368	Updated by this campaign
residual_harmonic_lstm_sequence_fw_sparse_rcim	te_residual_harmonic_lstm_sequence_remote_Fw_sparse_rcim	0.003234	Updated by this campaign
residual_harmonic_lstm_sequence_bw_sparse_rcim	te_residual_harmonic_lstm_sequence_remote_Bw_sparse_rcim	0.003440	Updated by this campaign
residual_harmonic_lstm_sequence_dense240	te_residual_harmonic_lstm_sequence_remote_global_dense240	0.003473	Updated by this campaign
residual_harmonic_lstm_sequence_fw_dense240	te_residual_harmonic_lstm_sequence_remote_Fw_dense240	0.003262	Updated by this campaign
residual_harmonic_lstm_sequence_bw_dense240	te_residual_harmonic_lstm_sequence_remote_Bw_dense240	0.003605	Updated by this campaign
residual_harmonic_lstm_sequence_dense360	te_residual_harmonic_lstm_sequence_remote_global_dense360	0.003477	Updated by this campaign
residual_harmonic_lstm_sequence_fw_dense360	te_residual_harmonic_lstm_sequence_remote_Fw_dense360	0.003351	Updated by this campaign
residual_harmonic_lstm_sequence_bw_dense360	te_residual_harmonic_lstm_sequence_remote_Bw_dense360	0.003556	Updated by this campaign

The family registries for the new Wave 2C model surfaces were created or refreshed by this campaign. The program-level training registry did not move to Wave 2C because the Wave 2B periodic GRU backward-only model still has the lower scalar test MAE.

Track 2 Boundary

Track 2 was not run as part of this closeout. Under the campaign governance rule, optional Track 2 verification remains a separate operator-approved workflow with a repository-owned launcher that can run locally or with - Remote .

Because Wave 2C does not beat the existing Wave 2B scalar best, a full Track 2 refresh is not mandatory for accepting this closeout. If reviewed later, the forward-only sparse RCIM residual harmonic GRU is the only Wave 2C candidate that should be promoted into the optional verification queue first.

Closeout Decision

The campaign is complete and successful from an execution standpoint: all 18 runs completed, the leaderboard and best-run artifacts exist, and the family registries were refreshed.

From a modeling standpoint, Wave 2C is a completed comparison branch rather than a new best branch. The residual harmonic temporal structure remains available for future analysis, but the current project best should stay on the Wave 2B periodic recurrent family until official verification says otherwise.

Recommended Follow-Up

1. Keep the Wave 2C artifacts as a completed negative/neutral comparison branch.
2. Do not replace the current Wave 2B scalar best with the Wave 2C winner.
3. Run optional Track 2 only if visual curve behavior is worth inspecting despite the weaker scalar campaign result.